

EvoCompute: Problem Definition, Literature Review and Strategic Business Case

*Working draft, 5 Oct 2026. Every number is tagged to a numbered source (Section 11) or labelled as an assumption. Evidence grades: **E1** = measured or published by a third party; **E2** = our model from stated assumptions; **E3** = hypothesis still to be tested.*

0. Executive summary

The idea in one line. EvoCompute is a RISC-V-class platform that senses its own health, workload and mission context, and switches between pre-certified hardware configurations under a hardware-enforced rule set, so that one piece of silicon serves several products and stays useful for longer.

What the evidence says (five findings):

- 1. The problem is real and growing.** Design cost rises steeply with node [2][3][4]. Only 14% of ASIC projects reach first-silicon success and 75% run late [7]. Silent data corruption from individual cores now shows up in hyperscale fleets [18][19]. Unplanned downtime costs the world's 500 largest firms about \$1.4T a year [10].
- 2. The idea is not new at the "self-aware chip" level.** Self-aware cyber-physical SoCs, hierarchical goal management and RL-supervised MPSoC control exist [21][22][23], as do self-healing RISC-V soft cores [25][26][27]. A claim of "first" will not survive review. The defensible novelty is the **unification** (workload + health + mission + security in one loop), a **hardware-enforced admissible-configuration set**, a **switching-cost model**, and **measured evidence across heterogeneous missions**. We add one new axis, **lifetime-carbon as an optimisation objective** (Section 8).
- 3. The strongest business argument is not energy.** At 15% energy saving, a 20 W node saves about \$2.6 a year (E2). Avoided stoppages are worth 1–3 orders of magnitude more per node, and the break-even rate of compute-attributable stoppages is tiny (Section 3.4). The catch is attribution: you must show the platform, not the plant, caused the avoided minutes.
- 4. The platform economics are conditional, not automatic.** Net NRE saving is positive only if EvoCompute avoids enough derivative designs relative to its own extra NRE (break-even rule in 3.3). As a standalone licensing business, the base case is **NPV-negative (about −\$6.8M at 15%)**; only the optimistic case clears a venture hurdle (IRR about 32%). Position it as a platform/research programme backed by public design incentives (India's Semicon 2.0 [38]), not as a stand-alone IP startup.
- 5. Verification is the existential risk.** Adaptivity multiplies the state space, and verification already takes 60–70% of chip-project effort [8]. The "certified configuration set" (the controller can only choose pre-verified states) is therefore the central design principle, not a feature.

Audit of numbers from the earlier thread

Claim in earlier thread	Status after checking	Use in deck?
Design cost \$298M at 7nm, \$540M at 5nm	These are IBS 2018 figures [3]. Critics put IBS 50–66% above real-world [4]; SemiEngineering suggested about \$160M / \$280M [3]; a real-world AI-chip estimate is \$62M at 7nm [4]. The IBS 2022 model reproduced in Arm's filing gives \$48M at 28nm, \$449M at 5nm [2]	Re-base to \$48M (28nm) and show a low-case
Learning-based DVFS saves 33% energy "(2024 IEEE)"	Verified, but the source is DVFO: arXiv 2023 / IPSN 2023 poster, tested on NVIDIA Jetson edge devices against other schemes, not custom silicon [24]	Cite as precedent , not as our target
"2026 IEEE TNS self-healing RISC-V: 95.34% / 99.94% recovery"	Could not locate. Related papers found: 45.09% effective software recovery under neutron irradiation [25]; hypervisor + partial reconfiguration for MPSoCs, 2025 [26]	Do not cite until the paper is found
Siemens: \$1.4T/yr, 11% of revenue, auto \$2.3M/h, 27 h/month	Verified [10][11]	Yes
IEA: data-centre demand to about 950 TWh by 2030	IEA says about 945 TWh, up from 415 TWh in 2024 [12][13]	Yes (945)
McKinsey: zonal E/E cuts material cost 10–20%	McKinsey states wiring harness is about 20% of E/E budgets and projects 18% zonal share by 2030 [32]; the 10%/20% cost figures appear on Bosch's page [33]	Attribute to Bosch , not McKinsey
Arm: up to 12 months faster, "millions" NRE	Arm now says "tens of millions" NRE saved and up to 12 months [5] (vendor claim)	Yes, as a competitor benchmark
ABB 2025 survey, IBM 18–31% predictive-maintenance saving, 2026 approximate-computing and 2025 timing-prediction papers	Not re-verified this session	Verify before use

1. Problem definition

1.1 Problem statement (SCQA)

Situation. Silicon is a bet placed 2–5 years before deployment, on a fixed architecture, with fixed safety margins, for an assumed workload and environment. The bet is getting more expensive: design cost at 28nm is about \$48M and at 5nm about \$449M on IBS's model [2].

Complication. Reality diverges from the bet in eight measurable ways:

#	Sub-problem	What goes wrong	Evidence
P1	Workload–silicon mismatch	Hardware is sized for peak or an assumed mix; resources idle or throttle	Dark/dim silicon, power-limited scaling [20]; energy-proportionality gap [42]
P2	Static margins	Worst-case voltage/timing guard-bands waste energy in typical conditions	Razor-style margin removal precedent [43]; learning-based DVFS shows double-digit savings [24]
P3	Silent and latent faults	Aging, marginal cores and radiation cause wrong answers without crashing	"A few mercurial cores per several thousand machines" at Google [18]; Meta fleet findings [19]
P4	Product fragmentation	Each market variant needs another design, verification and tape-out	Design cost [2][3][4]; Arm CSS exists precisely to cut this [5]
P5	Short effective life, high embodied carbon	Chips are retired for workload obsolescence or partial failure while their manufacturing carbon is sunk	Manufacturing dominates mobile and data-centre footprints [16]; 62 Mt e-waste in 2022, 22.3% formally recycled [17]
P6	Unverifiable adaptivity	A chip that changes behaviour multiplies the verification state space	14% first-silicon success; verification is 60–70% of effort [7][8]
P7	Adaptivity as an attack surface	Power/frequency control interfaces have been exploited	CLKSCREW fault attacks via DVFS [28]; Hertzbleed frequency side channel [29]
P8	Economic and regulatory friction	Buyers pay for certainty; sector regulation lags	Safety standards, CRA timelines [39]; limited DLI uptake in India [38]

Question. Can a single platform adapt to workload, health and mission **safely, provably and economically** enough to beat both static designs and point-solutions (DVFS-only, ECC/TMR-only, chiplets, FPGAs), and does that reduce lifecycle carbon?

1.2 Scope and definitions

- **In scope:** coarse-grained, pre-certified adaptation (frequency/voltage, cache ways, accelerator on/off, precision modes, redundancy level, fault isolation, task migration); an on-chip policy engine; a hardware constitution; a fault, thermal and workload evaluation on FPGA, then a test chip.
- **Out of scope for generation 1:** fine-grained runtime fabric reconfiguration, free-form online learning of safety-critical behaviour, certification in automotive/medical, multi-chip "EvoMesh" (kept as vision).
- **Definitions:**
 - *Mission profile:* a weighted objective vector (performance, energy, reliability, safety, security, latency, thermal, lifetime).
 - *Certified configuration set (CCS):* the finite set of hardware states verified offline; the policy engine may select only from it.
 - *Constitution:* immutable hardware invariants (for example $T < T_{max}$, safety core never disabled, memory isolation intact) that no policy can override.
 - *Switching economics:* reconfigure only if predicted benefit over the dwell time exceeds switching cost, with hysteresis and minimum residency.

1.3 Hypotheses and how each could be proved wrong

ID	Hypothesis	Metric	Target (design target, not claim)	Falsified if
H1	Unified control beats best single-axis control (DVFS-only, fault-only)	Energy, performance/W, recovery rate vs baselines B0–B3	$\geq 15\%$ energy saving and $\geq 25\%$ perf/W on dynamic workloads	Gain over B1 (DVFS-only) is $< 5\%$
H2	Adaptation overhead can stay small	Area overhead of sensing + policy + constitution	$\leq 10\%$	$> 15\%$ after synthesis
H3	Faults are survivable with graceful degradation	Recovered fraction, critical-function survival, silent-corruption rate	$\geq 95\%$ recovery within a stated fault model; zero silent corruption in critical tasks	Silent corruption observed in critical path
H4	One silicon can serve ≥ 4 mission profiles	Same RTL, different policies, each hitting its own objective	4 profiles each within 10% of a per-mission static optimum	Any profile $> 25\%$ worse than its static optimum
H5	The CCS keeps verification tractable	Number of certified states, formal-property coverage, verification effort	$\leq$ a stated cap on states; all invariants proven	Proof of constitution fails or effort exceeds the cap
H6	Adaptation is not exploitable to force unsafe states	Attack campaign on sensor, telemetry and policy inputs	Zero constitution violations; bounded availability loss	Any violation under attack
H7	Lifetime-aware policies extend useful service life	Years to a defined capability floor under injected aging	$\geq 30\%$ longer than static (assumption)	$< 10\%$ extension

Non-negotiable honesty rule for any presentation: report these as targets until measured, and report the failures too.

2. Literature review

2.1 Thematic map

Theme	Key sources	What they establish	Gap for EvoCompute
A. Why static silicon is straining	Esmailzadeh et al., ISCA 2011 [20]; Hennessy & Patterson 2019 [41]; Barroso & Hölzle 2007 [42]	Power-limited scaling, dark/dim silicon, need for specialisation; machines are inefficient at low utilisation	Specialisation fixes efficiency but freezes the bet; adaptation is the complement
B. Autonomic and self-aware computing	Kephart & Chess 2003 [40]; Dutt, Jantsch, Sarma 2016 [44]; Gonzalez-Martinez et al., 2024 survey [21]; CPSoC goal-management project [22]	Observe-decide-act loops; self-aware cyber-physical SoCs; two decades of MPSoC management over power, thermal, faults and security	Closest prior art. Mission/goal management is already studied. Our edge must be hardware-enforced safety, certified states, switching economics and measured multi-mission evidence
C. Learning-based power management	Zhang et al. (DVFO) [24]; Maurer et al., DATE 2020 [23]	RL/learned DVFS reports 33% average energy saving in an edge-cloud setting [24]; hierarchical supervisor + RL on an FPGA prototype obeys power limits [23]	Single-axis, or no formal guarantee on safety invariants
D. Fault tolerance and self-healing RISC-V	Bolchini & Miele 2014 [27]; Santos et al., TNS 2024 [25]; Cano-Páez et al., TNS 2025 [26]	Replication + partial reconfiguration + checkpoint/rollback can detect and repair upsets; 45.09% effective software recovery in neutron tests [25]	Reliability-only; no joint trade-off with energy and mission
E. Silent data corruption at scale	Hochschild et al. (Google) [18]; Meta fleet work [19]	Marginal cores produce wrong results silently; a few per several thousand machines; detection and isolation needed	Strong motivation for on-chip health sensing and core isolation
F. Commercial adaptive platforms	Microchip RT PolarFire SoC [30]; AMD Versal adaptive SoC [31]; Arm CSS [5]	RISC-V + FPGA fabric with radiation-tolerant variant; functional-safety-certified toolchain (ISO 26262 ASIL-D / IEC 61508 SIL-3) [30]; adaptive SoCs for space [31]; pre-integrated subsystems cut time and NRE [5]	They adapt <i>function</i> (fabric) or <i>integration</i> ; none exposes a mission-level, closed-loop, constitution-bounded policy
G. Security of adaptive control	Tang et al., CLKSCREW [28]; Wang et al., Hertzbleed [29]	DVFS interfaces enabled fault injection into TrustZone and a remote frequency side channel	The policy engine itself must be hardened; telemetry must be authenticated
H. Verification and safety economics	Wilson Research / Siemens 2024 [7][8][9]	14% first-silicon success; 75% projects behind schedule; verification 60–70% of effort	Defines the cost of adaptivity; justifies the certified-state approach
I. Sustainability and LCA	BCG/SEMI [14]; imec.netzero [15]; Gupta et al. [16]; UN E-waste Monitor [17]; IEA [12]	Lifecycle split of device emissions; fab electricity dominates fab footprint; manufacturing dominates many compute footprints; e-waste growing faster than recycling	No hardware-architecture work that optimises lifetime-carbon at runtime (to our knowledge; to be confirmed by a systematic search)

Theme	Key sources	What they establish	Gap for EvoCompute
J. Design economics and reuse	IBS via Arm filing [2]; SemiEngineering [3]; Silicon Analysts [4]; Arm CSS [5][6]	Cost by node; headline numbers likely inflated; reuse via compute subsystems claims up to 12 months and tens of millions saved	Reuse via runtime adaptation is untested against reuse via pre-integration

2.2 Source-by-source notes (for the related-work section)

- **Esmailzadeh et al. 2011 [20]**. Shows multicore scaling is power-limited "to a degree not widely appreciated" and introduces the dark-silicon framing. Their retrospective says real chips show "dim" silicon with aggressive voltage/frequency scaling, which is exactly the regime where intelligent management pays. *Use*: motivation, not a number.
- **Gonzalez-Martinez et al. 2024 [21]**. Surveys two decades of MPSoC management and describes autonomous management that integrates self-awareness of environment, behaviour and objectives. *Use*: the anchor survey; our related-work section must position against it.
- **CPSoC / hierarchical goal management [22]**. Publications on goal-driven autonomy, hierarchical dynamic goal management and self-optimising learning with self-adaptive control. *Implication*: "Mission → architecture" in the earlier thread is **not new as a concept**. Credit it, then differentiate.
- **Maurer et al. DATE 2020 [23]**. Supervisor + RL identifies optimised MPSoC operating parameters at runtime while strictly obeying power constraints, on an FPGA prototype. *Implication*: a constraint-respecting learned controller already exists for power; ours extends the constraint set to safety/security invariants with a certified-state design.
- **DVFO [24]**. 33% average energy reduction and 28.6–59.1% latency reduction on edge devices, within about 1% accuracy loss. *Caveat*: different platform and baselines than ours; treat as an upper-plausibility marker.
- **Santos et al. 2024 [25]**. Hybrid hardening of a RISC-V SoC with checkpoint/rollback; 45.09% effective software recovery in neutron irradiation with low overhead. *Use*: a measured, modest baseline that shows fault recovery is not trivially near-100%.
- **Hochschild et al. 2021 [18] and Meta [19]**. Mercurial cores are rare but visible at fleet scale and corrupt even encryption. *Use*: the best argument that health awareness must be in the core, not only in fleet software.
- **CLKSCREW [28] and Hertzbleed [29]**. Prove that energy-management interfaces are an attack surface. *Use*: the justification for authenticated policy updates, rate-limited actuation, and a constitution.
- **Wilson Research 2024 [7][8][9]**. Verification effort grows faster than design complexity; first-silicon success was about 30% in 2012 and 14% in 2024. *Use*: the number a sceptical reviewer will cite against us; our answer is H5.
- **Gupta et al. [16] vs BCG/SEMI [14]**. They disagree on where the footprint sits. SEMI/BCG attribute 63% of 2021 device lifetime emissions to **use**, 21% to manufacturing, 16% to supply chain [14]; Gupta et al. find most emissions from mobile and data-centre equipment come from **manufacturing and infrastructure** [16]. The difference is scope and boundary. *Implication*: our sustainability claim must hold under **both** views: energy saving covers the use-dominated view, lifetime extension covers the manufacturing-dominated view.

2.3 Novelty position (what we can and cannot claim)

Claim	Defensible?	Why
"First self-aware or self-healing RISC-V"	No	Prior art in B and D
"Mission-driven management of an SoC"	Partly	Goal management exists [22][23]; differentiate on hardware enforcement and certified states
"Unified workload + health + security + mission loop with hardware constitution, measured across ≥ 4 missions"	Yes, if measured	No single published system found doing all of it (to be confirmed by a systematic search before submission)
"Switching-economics-aware policy (benefit over dwell time > switching cost)"	Likely	Treat as a core contribution; test against hysteresis-only baselines
"Lifetime-carbon-aware runtime policy"	Likely novel	No direct precedent found; needs a proper search

Search protocol to close the gap (do before any novelty claim): IEEE Xplore, ACM DL and arXiv with strings combining {self-aware | autonomic | adaptive} × {SoC | RISC-V | MPSoC} × {fault | thermal | mission | goal | carbon}; backward and forward citation chaining from [21], [22], [23], [25]; record hits and exclusions in a PRISMA-style table.

3. Market and financial analysis

3.1 Market sizing (E1 inputs, E2 derivation)

Layer	Size	Source and note
Global semiconductor market	H1 2026 \$702B (+102% YoY); 2026 ≈ \$1.66T; 2027 ≈ \$2.1T [34]	Memory-driven surge (memory +305% in H1); not a relevant TAM for this idea
TAM: industrial ICs	\$65.5B in 2026 → \$89.8B in 2031 (6.5% CAGR) [35]; another estimate \$63.5B in 2026 [36]	Includes analog, power, sensors
Microcontrollers within industrial ICs	22.95% of the market by 2031, ≈ \$20.6B [35]	Programmable compute subset
Edge-AI chips	\$20.6B (2026) [37a] vs \$29.9B (2026) [37b]	Vendor definitions differ widely; overlaps with industrial
SAM (our estimate)	≈ \$13–25B in 2026	Industrial programmable compute (MCU/MPU/edge processors for robots, drones, industrial edge) at a similar share of today's industrial ICs, plus the industrial edge-AI portion; overlap not fully removed
SOM	0.1–0.5% of SAM by year 8–10 ≈ \$15–125M/yr of silicon value influenced (E2, assumption)	Only meaningful as value-influenced revenue; vendor-side revenue is smaller (3.3)

Reading: the opportunity is a niche inside a large market. That is fine for a research-grade UDT story but means the pitch should not lean on "trillion-dollar market" framing.

3.2 Three lenses on economics

Lens	Who	Question
A. Platform vendor	Company building and licensing EvoCompute	Does the business earn more than its cost of capital?
B. Product developer	Fabless firm or OEM needing several variants	Does one adaptive platform cost less than several fixed derivatives?
C. Customer (end-user TCO)	Plant or fleet operator	Does the value per node exceed the extra cost per node?

3.3 Lens B: net NRE saving (28nm, E2)

Assumptions: a three-product portfolio. Conventional route: one full design F plus two derivatives at $d \times F$ each. EvoCompute route: one platform at $F(1 + e)$ plus $m \times F$ per added mission profile. Full design cost $F = \$48\text{M}$ (IBS via Arm filing [2]) and a lower realistic case \$25M (our adjustment in light of [3][4]). Per-profile policy and evidence cost $m = 4\%$ of F .

Net NRE saving = $(2d - e - 2m) \times F$. Break-even extra-NRE share: $e^* = 2d - 2m$.

Net saving (\$M), $m = 4\%$	$e = 15\%$	$e = 30\%$	$e = 50\%$
$F = \$48\text{M}, d = 25\%$	13.0	5.8	-3.8
$F = \$48\text{M}, d = 35\%$	22.6	15.4	5.8
$F = \$48\text{M}, d = 50\%$	37.0	29.8	20.2
$F = \$25\text{M}, d = 25\%$	6.7	3.0	-2.0
$F = \$25\text{M}, d = 35\%$	11.7	8.0	3.0
$F = \$25\text{M}, d = 50\%$	19.2	15.5	10.5

Reading: with $d = 35\%$ and $m = 4\%$, break-even $e^* \approx 62\%$, so there is headroom, but the saving turns negative when derivatives are cheap ($d = 25\%$) and EvoCompute's extra NRE is high ($e = 50\%$). **Given the verification evidence [7][8], a realistic e is 30–50%, not 15%.** The earlier thread's "\$253M saved" case (one fully avoided \$298M design) should not be used.

Unit economics (E2): extra unit cost = area overhead $\times$ die cost + extra test. With a \$9 die: 5% area and \$0.10 test = \$0.55; 10% and \$0.20 = **\$1.10**; 15% on a \$12 die and \$0.30 test = \$2.10.

3.4 Lens C: customer value per node (E2)

Annual value per node = $\lambda \times \mu \times T \times C$, where λ = compute-attributable stoppage events per node-year, μ = fraction mitigated by graceful degradation or recovery (assume 0.5), T = hours per event (assume 0.5), C = cost per hour.

C (source)	$\lambda = 0.005$ (1 per 200 nodes)	$\lambda = 0.02$ (1 per 50 nodes)
\$36k/h, FMCG low end [11]	\$45	\$180
\$260k/h, cross-sector average cited secondhand (Aberdeen)	\$325	\$1,300
\$2.3M/h, automotive [10]	\$2,875	\$11,500

Energy comparison: 15% off a 20 W node = 3 W = 26.3 kWh/yr = **\$2.63/yr at \$0.10/kWh** (131 kWh over 5 years). Energy is an order of magnitude smaller than avoided downtime unless λ is extremely small.

Break-even stoppage rate. If the extra cost per node (silicon \$1.10 plus about \$5 adoption cost, our assumption) is amortised over 5 years (about \$1.22/yr), EvoCompute pays back when $\lambda \times \mu \times T \times C \geq \1.22 , i.e. $\lambda \approx 1.4 \times 10^{-4}$ events/node-year at $C = \$36\text{k/h}$, 1.9×10^{-5} at $\$260\text{k/h}$, and 2.1×10^{-6} at $\$2.3\text{M/h}$. In words: the platform needs to prevent a plant-stopping compute fault roughly once per 7,000 node-years at the lowest cost level.

Critical caveat (attribution). The plant stops only if the compute fault was the cause and the stoppage would otherwise have happened. Siemens' figures are for all unplanned downtime [10]; the share caused by compute-node faults is unknown and must be measured with a design partner. Do not claim the Siemens totals as EvoCompute savings.

3.5 Lens A: vendor model (10-year, 15% discount rate, E2)

Assumptions: 3 years of development (\$4–5M/yr), then \$3M/yr operating cost; licence fee per licensee; licensees ship units from year 2 after signing (50% in the first year); royalty per unit. All figures are assumptions to be replaced with partner quotes.

Scenario	Dev \$M/yr	New licensees (yrs 3–7)	Fee \$M	Royalty \$/unit	Units/licensee/yr (M)	NPV @15%	IRR	Payback
Pessimistic	5.0	1,1,2,2,2	1.0	0.10	0.2	– \$18.6M	n/a	none
Base	4.0	2,3,4,4,4	1.5	0.20	0.3	–\$6.8M	n/a	none in 10 yrs
Optimistic	4.0	3,5,6,7,7	2.0	0.30	0.5	+\$8.4M	31.6%	4.3 yrs

Sweep on the base case: raising royalty alone to \$1.00/unit still gives NPV –\$3.8M (IRR 5.3%); raising volume alone to 1.5M units per licensee gives the same. A combination of fee \$2.0M, royalty \$0.50 and 0.6M units per licensee gives NPV +\$0.3M (IRR 15.7%, payback 5.4 years).

Reading: a stand-alone licensing business does **not** clear a 15% hurdle without optimistic volume and pricing. The honest framing is a **platform and research programme** funded by grants and design-linked incentives, with services (policy tuning, evidence packs) and later IP licensing. This is the finding most likely to surprise a reviewer, and it makes the paper more credible.

3.6 Energy-at-scale envelope (not a claim)

If adaptive hardware influenced 5% of data-centre electricity and cut it 10%, using IEA's 945 TWh for 2030 [12]: $945 \times 5\% \times 10\% = 4.7 \text{ TWh/yr}$, about \$470M at \$0.10/kWh, or roughly 1.9–3.3 Mt CO₂ at an assumed 0.4–0.7 kg/kWh grid factor. Present only as an *opportunity envelope*. Data centres already run mature schedulers and DVFS, so the real bar is high.

4. Industry impact

4.1 Value chain: where EvoCompute touches

Stage	Today	Effect of EvoCompute	Who gains / loses
EDA and IP	Tools verify fixed designs	Needs mode-aware verification, formal proof of invariants	EDA vendors gain a new verification market; IP vendors must expose health/telemetry hooks
Design house	One design per market variant	Fewer variants, more policy/software work	Gains if H4 holds; loses derivative-design revenue
Foundry / OSAT	Wafer and package volume per SKU	Possibly larger die (area overhead), fewer SKUs	Slightly more silicon per unit; less SKU fragmentation
OEM / integrator	Qualifies hardware per product	Qualifies a platform plus mission profiles	Lower BOM variety; higher qualification burden for the first profile
Operator	Replaces parts on failure or obsolescence	Graceful degradation, health telemetry, longer service	Fewer unplanned stops; needs trust in autonomous adaptation
Recycler / policy	Receives retired hardware	Fewer, later retirements	Lower e-waste per service-year [17]

4.2 Competitive landscape

Alternative	Strength	Weakness vs EvoCompute's pitch	Evidence
Fixed ASIC	Best efficiency for a known workload	Frozen; high NRE [2]	E1
Arm Compute Subsystems / chiplets	Pre-integrated, verified; up to 12 months faster, tens of millions NRE saved [5]	Statically composed; directly competes for the "avoid redesign" argument	E1 (vendor claim)
FPGA / SoC FPGA (Microchip PolarFire SoC, AMD Versal)	Reconfigurable; RISC-V + fabric; radiation-tolerant variant; certified toolchain [30][31]	Higher power/cost for fixed functions; adaptation of <i>function</i> , not a closed-loop mission policy	E1
Classic DVFS / power management	Mature, cheap	Single-axis; also an attack surface [28][29]	E1
Fault-tolerant cores (TMR/ECC/DPR)	Proven for space and safety [25][26][27]	Always-on overhead (replicated logic under TMR typically costs about 3× area); reliability-only	E1 / E2
Self-aware MPSoC research	Strong theory [21][22][23]	Mostly simulation or FPGA, no hardware-enforced certified states, limited multi-mission evidence	E1

Porter's Five Forces (summary):

Force	Intensity	Reasoning
Rivalry	High	Arm, Microchip, AMD, Intel/Altera, RISC-V IP houses, plus open-source cores
Threat of substitutes	High	CSS/chiplets, FPGAs and software-defined orchestration solve overlapping problems
Buyer power	High	OEMs buy on qualified, certified, second-sourced parts
Supplier power	Medium	Foundry and EDA concentrated; open-source flows (OpenROAD/OpenLane) reduce dependence for prototypes
Threat of entry	Medium	Open RISC-V lowers core barriers; verification, safety evidence and trust are the real barriers

Likely incumbent response (hypothesis): add health telemetry and DVFS policy engines to existing SoCs, and market "predictive reliability" features. Our durable differentiators are therefore the **open, auditable, constitution-bounded policy** and the **evidence base**, not the existence of an adaptive loop.

5. Friction and feasibility

5.1 Barriers

Type	Barrier	Severity	Mitigation
Technical	State-space explosion: 4 modes × 3 voltages × 3 frequencies × 3 cache modes × 2 accelerator states = 216 configurations before corners and fault states (earlier thread)	High	Certified configuration set; formal proof of invariants; test only reachable-state transitions
Technical	Adaptation overhead can erase gains on short workloads	Medium	Switching-economics model; hysteresis; minimum residency; benchmark against static
Technical	Control oscillation	Medium	Dwell-time and confidence thresholds; stability analysis
Technical	Area overhead	Medium	Coarse-grained morphing only; hard cap at ≤10%, kill at >15%
Verification	14% first-silicon success; 75% projects behind schedule [7]	High	Reuse proven RISC-V cores; formal-first; FPGA before silicon
Security	DVFS-style interfaces exploited before [28][29]	High	Authenticated policy updates, rate-limited actuation, redundant sensors, immutable safe-state controller
Regulatory	Safety standards (IEC 61508, ISO 26262) demand predictable behaviour; vendor toolchains already carry ASIL-D/SIL-3 certificates [30]	High for safety-critical, low for non-safety supervisory nodes	Start in non-safety-critical industrial vision/supervision; design evidence for a later safety case
Regulatory	EU CRA: reporting of actively exploited vulnerabilities applies since 11 Sep 2026, full requirements from 11 Dec 2027; sector-regulated areas (for example automotive, medical) are exempt [39]	Medium	Build vulnerability handling and secure update into the design; treat CRA as a market-access requirement for connected industrial products
Cultural	Engineers distrust "AI in the chip"; operators prefer determinism	High	"AI proposes, constitution decides"; expose logs and replay; publish negative results
Economic	Incumbent reuse offerings already promise NRE savings [5]	High	Compete on lifetime, resilience and mission flexibility, not only NRE
Ecosystem (India)	DLI uptake has been limited and fund utilisation slow [38]; Semicon 2.0 (approved 15 Jul 2026) adds deployment-linked incentives to offset tape-out cost [38]	Medium	Align the programme with Semicon 2.0 design, IP and productisation priorities

5.2 Roadmap with stage gates

Phase	Months	Deliverable	Gate (continue only if)
0. Golden model + FPGA prototype	0–3	RISC-V core + sensors, 4 modes, fault injector, baselines B0 (static), B1 (DVFS-only), B2 (workload-aware), B3 (fault-tolerant-only) on an FPGA board	Beats B1 on at least one workload mix by >5% or shows a recovery benefit
1. Policy + constitution	3–9	Table/bandit policy, then tiny MLP; constitution as hardware checks; CCS defined; formal proofs of invariants	All invariants proven; area overhead $\leq 15\%$ post-synthesis
2. Attack and fault campaigns	6–12	Thermal, timing, SEU, memory corruption and telemetry-spoofing campaigns	Zero constitution violations under attack
3. Test chip	9–18	MPW tape-out on an open or institutional node; silicon characterisation	Measured overhead and energy within 20% of FPGA/synthesis projections
4. Pilot	18–36	Industrial vision or UAV pilot with a design partner; measure λ (compute-attributable stoppage rate)	Partner data support λ above break-even (3.4)
5. Productisation	36+	28nm design, safety-case preparation, licensing/services	Business case clears hurdle (3.5) with partner-validated inputs

Reuse note: an existing RV32 core flow and an FPGA board with a working Quartus/OpenLane toolchain can compress Phase 0 substantially.

Critical success factors: (1) a reproducible, open evaluation harness with baselines; (2) formal proofs of the constitution; (3) one credible design partner for λ data; (4) honest reporting of overhead and failures; (5) a clear first market (non-safety-critical industrial edge).

Kill criteria: area overhead > 15% after synthesis; gain over DVFS-only < 5% on realistic workloads; any constitution violation under attack; verification effort for the CCS beyond the stated cap.

6. Value proposition and selling points

Unique selling proposition: "One silicon, many optimal machines, safe by construction."

Stakeholder	Benefit	Evidence now	Evidence needed
Plant / fleet operator	Fewer plant-stopping compute faults, health telemetry, longer service life	E1 context [10][11], E2 break-even (3.4)	Measured λ and μ
OEM / product developer	Fewer variants, one qualification base	E2 (3.3)	Partner NRE data
Chip designer / EDA	Reusable verified platform, open policy interface	E3	Prototype
Policy maker	Design-led, indigenous, sustainable computing story aligned with Semicon 2.0 [38]	E1 policy context	Pilot outcomes
Sustainability officer	Lower use-phase energy and lower carbon per service-year	E2 envelopes (Section 8)	LCA with manufacturer data
Security officer	Bounded, auditable adaptation	E3	Attack campaign

ESG alignment: use-phase energy (63% of 2021 device lifetime emissions in SEMI/BCG's accounting [14]), embodied carbon amortisation (manufacturing-dominant view [16]), e-waste reduction [17], and resilience. Frame all of these as **potential**, with measured values to follow.

7. Pros, cons, SWOT

7.1 Pros and cons with quantification

Pros	Size (grade)	Cons	Size (grade)
Avoided derivative NRE	\$3–37M per three-product portfolio at 28nm (E2, table 3.3)	Extra platform NRE, verification, policy evidence	e = 15–50% of F; plausible 30–50% (E2/E3)
Avoided stoppage value	\$45–\$11,500 per node-year depending on inputs (E2)	Extra unit cost	\$0.55–\$2.10 per unit (E2)
Energy saving	\$2.6/yr per 20 W node at 15% (E2); 33% precedent in a different setting [24]	Adaptation overhead, oscillation risk	Workload-dependent (E3)
Longer useful life	30% longer life cuts amortised embodied carbon about 23% (E2 arithmetic)	Adoption: buyers prefer certified, deterministic parts	Qualitative (E1 context [7])
Fault resilience	Recovery benefit vs static (to be measured)	Security exposure of adaptive interfaces	Real precedents [28] [29]
Strong fit with VLSID theme and Semicon 2.0	Qualitative	Crowded prior art	See 2.3

Do the pros outweigh the cons? At *research level*, yes: the downside is bounded (a prototype) and the upside includes publishable negative or positive results. At *commercial level*, **not yet proven**: the base-case vendor NPV is negative (3.5) and incumbents offer overlapping savings (4.2). The decision therefore rests on three numbers we do not have: λ , e, and measured energy gain. The roadmap is built to get them cheaply.

7.2 SWOT

Strengths	Weaknesses
Unified loop plus constitution; open, auditable design; buildable on FPGA	Small team and limited silicon access; no field data yet
Clear sustainability angle (lifetime-carbon objective)	Added NRE and verification burden
Opportunities	Threats
Semicon 2.0 design incentives [38]; CRA pushes secure, updateable hardware [39]; rising fleet-scale SDC awareness [18][19]	Arm CSS and FPGA vendors already address reuse and reliability [5][30][31]; novelty challenges from self-aware SoC literature [21][22][23]

7.3 Risk register (top items)

Risk	Likelihood	Impact	Mitigation	Owner/Gate
Gain over DVFS-only is small	Medium	High	Early baseline comparison; pivot to reliability + lifetime story	Gate 0
Verification cost exceeds budget	High	High	CCS cap; formal-first	Gate 1
Policy exploited by attacker	Medium	High	Authenticated updates, safe-state controller	Gate 2
Reviewer rejects novelty	Medium	High	Systematic search; explicit differentiation (2.3)	Before submission
No partner data for λ	Medium	Medium	Public failure datasets; synthetic fault models stated clearly	Gate 4

8. Sustainability and vision

8.1 What the sustainability case rests on

- 1. Use-phase energy.** In SEMI/BCG's accounting, 63% of lifetime emissions from 2021-produced devices (500 Mt CO₂e) come from device use, 21% from manufacturing, 16% from the supply chain [14]. EvoCompute lowers use-phase energy where workloads vary.
- 2. Embodied carbon.** Manufacturing and infrastructure dominate many compute footprints [16]; fab electricity can reach 60% of a fab's footprint, and chips contribute about 30% of AI data-centre carbon in imec's framing [15]. A chip that stays useful longer amortises sunk carbon over more service-years.
- 3. Circularity.** E-waste reached 62 Mt in 2022 with 22.3% formally recycled and is on course for 82 Mt by 2030 [17]. Longer service life and graceful degradation reduce retirements.

8.2 Envelope calculations (E2, assumptions explicit)

Using SEMI/BCG's 500 Mt baseline: embodied (supply chain + manufacturing) = 37% = **185 Mt**; use = 63% = **315 Mt** [14]. If EvoCompute-class platforms influenced a share s of devices:

Share s	Lifetime +30% → embodied amortisation falls 23.1%	Energy –15% on use phase	Combined (Mt CO ₂ e per year-equivalent)
0.5%	0.21	0.24	0.45
1%	0.43	0.47	0.90
2%	0.85	0.94	1.79

These are *illustrative upper-bound arithmetic*, assuming the lifetime and energy effects materialise and are not double-counted. Their purpose is to show order of magnitude (sub-Mt to low Mt per year at 0.5–2% influence), not to forecast.

8.3 The new sustainability contribution: carbon-aware lifetime policy

Extend the earlier utility function with a lifetime-carbon term:

$$U = w_p \cdot P + w_e \cdot E + w_r \cdot R + w_s \cdot S - w_t \cdot T - w_c \cdot (\text{embodied carbon} \div \text{expected remaining service-years})$$

The policy engine then prefers configurations that slow wear (lower voltage, lower temperature, spare-unit rotation) when mission slack allows, extending the capability floor. **Test (H7):** under injected aging, compare years to a capability floor vs static. This turns the sustainability story from a slogan into a measurable result.

8.4 Vision: a better silicon future

- **Today:** design a fixed chip for a guessed future, retire it when the guess fails or one block ages out.
- **With EvoCompute-class platforms:** a verified family of states, chosen at run time by a bounded policy, with health that is observable and reportable, and with service life managed as a resource.
- **For policy makers:** programmes such as Semicon 2.0 can fund *evidence* (open benchmarks, shared fault datasets, safety-case templates), not only fabs and tape-outs; and procurement can reward lifetime-carbon reporting for connected industrial equipment.

9. Solution map: every defined problem to a mechanism and a test

Problem	Mechanism in EvoCompute	Metric	Experiment	Residual risk
P1 Workload mismatch	Resource trading across CPU/accelerator/DSP modes; predictive workload horizon	Perf/W, utilisation	Mixed workload traces vs B0–B3	Gain may be small vs DVFS-only
P2 Static margins	Health-aware DVFS within constitution; timing-margin sensing	Energy at iso-performance	Sweep V/F under thermal stress	Margin recovery limited by sensor accuracy
P3 Silent/latent faults	Parity/ECC/lockstep sampling, anomaly detection, core isolation, rollback	Recovery %, silent corruption rate	SEU, memory, timing-violation injection	Fault model coverage
P4 Fragmentation	Mission profiles as policy + CCS subset	Profiles served within 10% of static optimum	≥4 profiles on the same RTL	NRE e may exceed break-even (3.3)
P5 Short life / embodied carbon	Lifetime-carbon utility term; wear-levelling across spare units	Years to capability floor; carbon per service-year	Aging injection, LCA arithmetic	Life extension unproven until measured
P6 Unverifiable adaptivity	CCS + formal invariants + constitution + replayable logs	States certified, properties proven, effort	SymbiYosys/formal on invariants; coverage report	Verification cost (60–70% norm [8])
P7 Attack surface	Authenticated policy updates, rate limiting, redundant sensors, safe-state	Violations under attack	Telemetry-spoofing and fault-attack campaigns [28][29]	Novel attacks
P8 Economic/regulatory friction	First market non-safety-critical; evidence pack; open policy interface; align with incentives [38][39]	λ vs break-even; partner LOI	Pilot	Incumbent response [5] [30]

10. What to validate next (in priority order)

1. **Systematic novelty search** (protocol in 2.3) before any claim of newness.
2. **Phase 0 baseline experiment**: does unified control beat DVFS-only by >5%? This one result decides whether the energy story survives.
3. **Measure λ** with a partner or from public failure datasets; if λ is far below the break-even in 3.4, drop the reliability-ROI pitch.
4. **Replace e** (extra NRE share) with a bottom-up estimate from the Phase 1 area and verification numbers.
5. **Locate or discard** the unverified claims in the audit table (self-healing TNS 2026 paper, IBM 18–31%, ABB survey, 2025–26 IEEE efficiency papers).
6. **Get an LCA figure** for the target node (imec.netzero [15] or manufacturer data) to replace the SEMI/BCG split in 8.2.

11. References

Retrieved 5 Oct 2026 unless marked "from memory". Vendor and market-research figures are secondary and should be re-checked against the primary document before they go on a slide.

Conference and policy

[1] VLSID 2027, *Call for User Design Track (UDT)*. <https://vlsid.org/call-for-udt/> [38] India: Semicon India Programme page (Semicon 2.0 approved 15 Jul 2026, ₹1,27,500 crore): <https://en.vikaspedia.in/viewcontent/e-governance/digital-india/semicon-india-program?lgn=en> ; YourStory, "Finance Ministry panel clears Rs 1.25 lakh crore outlay for ISM 2.0" (24 DLI projects, 105 firms with EDA access, 23 tapeouts): <https://yourstory.com/2026/06/finance-ministry-panel-1-25-lakh-crore-india-semiconductor-mission> ; IMPRI critique on DLI uptake: <https://www.impriindia.com/?p=74034> [39] EU Cyber Resilience Act timeline: <https://www.itemis.com/en/glossary/cyber-resilience-act/> ; <https://www.cyberresilienceact.eu/explained.html>

Design economics, verification, competitors

[2] AnySilicon, "How Much Does ASIC Design Cost in 2026?" (IBS model reproduced in Arm's 2023 filing: ~\$48M at 28nm, \$449M at 5nm, \$725M at 2nm). <https://any silicon.com/news/how-much-does-asic-design-cost-in-2026-real-price-ranges-by-process-node/> [3] Semiconductor Engineering, "What Will That Chip Cost?" <https://www.semiengineering.com/what-will-that-chip-cost/> [4] Silicon Analysts, Chip Design & NRE Cost by Node (IBS vs SemiAnalysis-adjusted). <https://siliconanalysts.com/market-data/nre-design-cost> [5] Arm Newsroom, "How Arm Compute Subsystems Accelerate the Future of Chip Design". <https://newsroom.arm.com/blog/arm-compute-subsystems-css-explainer> [6] Arm Newsroom, custom-silicon partnership post (2nm cost \$500–700M claim). <https://newsroom.arm.com/blog/ai-chip-design-partnership> [7] Siemens EDA / Wilson Research Group, *2024 IC/ASIC Functional Verification Trend Report*. <https://verificationacademy.com/topics/planning-measurement-and-analysis/wrg-industry-data-and-trends/2024-siemens-eda-and-wilson-research-group-ic-asic-functional-verification-trend-report/> [8] H. Foster, "Why First-Silicon Success Is Getting Harder for System Companies" (Siemens, 3 Sep 2025). <https://blogs.sw.siemens.com/verificationhorizons/2025/09/03/why-first-silicon-success-is-getting-harder-for-system-companies> [9] Embedded.com, interview on chip-verification challenges (first-silicon about 30% in 2012). <https://www.embedded.com/?p=4497797> [30] Microchip, RT PolarFire SoC FPGA: <https://microchip.com/en-us/products/fpgas-and-plds/radiation-tolerant-fpgas/rt-polarfire-soc> ; PolarFire family page (Libero certified ISO 26262 ASIL-D / IEC 61508 SIL-3): <https://microchip.com/en-us/products/fpgas-and-plds/fpgas/polarfire-fpgas> [31] ESA workshop session listing (AMD Versal adaptive SoC for space). <https://indico.esa.int/event/531/sessions/2097> [32] McKinsey, "Getting ready for next-generation E/E architecture with zonal compute". <https://mckinsey.com/industries/semiconductors/our-insights/getting-ready-for-next-generation-ee-architecture-with-zonal-compute> [33] Bosch Mobility, E/E architecture. <https://bosch-mobility.com/en/mobility-topics/ee-architecture/>

Markets

[34] WSTS Q2 2026 release: <https://www.wsts.org/esraCMS/extension/media/f/WST/7745/WSTS-Q2-Release-2026-08-06.pdf> ; Bits&Chips, 11 Aug 2026: <https://bits-chips.com/article/global-semiconductor-market-doubles-in-h1-2026/> [35] Mordor Intelligence, Industrial Integrated Circuits Market. <https://www.mordorintelligence.com/industry-reports/industrial-integrated-circuits-market> [36] Coherent Market Insights, Industrial Chips Market (via GII). <https://www.gii.tw/report/coh2058284->

industrial-chips-market-by-product-type-by.html [37a] Meticulous Research, Edge AI Chips Market. <https://www.meticulousresearch.com/product/edge-ai-chips-market-6869> [37b] Fortune Business Insights, Edge AI Semiconductor Market. <https://www.fortunebusinessinsights.com/edge-ai-semiconductor-market-117383>

Industrial cost of downtime, energy, sustainability

[10] Siemens, *The True Cost of Downtime 2024*. https://assets.new.siemens.com/siemens/assets/api/uuid:1b43afb5-2d07-47f7-9eb7-893fe7d0bc59/TCOD-2024_original.pdf [11] vsight.io summary of Siemens/Senseye (FMCG \$36k/h, 27 h/month, 25 incidents/month). <https://vsight.io/blog/cost-of-unplanned-downtime-manufacturing/> [12] IEA, *Energy and AI* (Apr 2025). <https://www.iea.org/news/ai-is-set-to-drive-surging-electricity-demand-from-data-centres-while-offering-the-potential-to-transform-how-the-energy-sector-works> [13] Scientific American on the IEA report (415 TWh in 2024; 945 TWh in 2030). <https://scientificamerican.com/article/ai-will-drive-doubling-of-data-center-energy-demand-by-2030> [14] BCG / SEMI Semiconductor Climate Consortium, *Transparency, Ambition and Collaboration* (2023): <https://discover.semi.org/rs/320-QBB-055/images/Transparency-Ambition-and-Collaboration-BCG-SEMI-SCC-20230919.pdf> ; summary (500 Mt; 16/21/63%): <https://www.3dincites.com/2023/10/tracking-and-reporting-sustainability-a-stake-in-the-sand/> [15] imec, "How can we reduce the environmental impact of chip manufacturing?" (imec.netzero). <https://www.imec-int.com/en/articles/how-can-we-reduce-environmental-impact-chip-manufacturing> [16] U. Gupta et al., "Chasing Carbon: The Elusive Environmental Footprint of Computing," HPCA 2021 / IEEE Micro 42(4), 2022. <https://arxiv.org/abs/2011.02839> [17] UNITAR/ITU, *Global E-waste Monitor 2024*. <https://www.unitar.org/about/news-stories/press/global-e-waste-monitor-2024-electronic-waste-rising-five-times-faster-documented-e-waste-recycling>

Architecture, self-awareness, reliability, security

[18] P. Hochschild et al., "Cores that don't count," HotOS 2021. <https://research.google/pubs/cores-that-dont-count/> [19] Meta Engineering, silent errors / silent data corruption at scale (H. Dixit et al., 2021). <https://engineering.fb.com/2022/03/17/production-engineering/silent-errors/> [20] H. Esmailzadeh et al., "Dark Silicon and the End of Multicore Scaling," ISCA 2011. <https://doi.org/10.1145/2000064.2000108> [21] G. Gonzalez-Martinez et al., "A Survey of MPSoC Management toward Self-Awareness," *Micromachines* 15(5):577, 2024. <https://doi.org/10.3390/mi15050577> [22] CORDIS project 705617, "Self-Aware CPSoCs with Hierarchical Goal Management" (publication list). <https://cordis.europa.eu/project/id/705617> [23] F. Maurer et al., hierarchical supervisor / reinforcement-learning MPSoC control, DATE 2020 special session. <https://date20.date-conference.com/node/627> [24] Z. Zhang et al., "DVFO: Learning-Based DVFS for Energy-Efficient Edge-Cloud Collaborative Inference," arXiv:2306.01811. <https://arxiv.org/abs/2306.01811> [25] D. A. Santos et al., "Hybrid Hardening Approach for a Fault-Tolerant RISC-V System-on-Chip," IEEE TNS 71(8), 2024. <https://doi.org/10.1109/TNS.2024.3406021> [26] J. Cano-Páez et al., "Architecture for error detection and recovery in MPSoCs: A hypervisor approach using Dynamic Partial Reconfiguration," IEEE TNS 2025. <https://doi.org/10.1109/TNS.2025.3534431> [27] C. Bolchini, A. Miele, "Design and implementation of a self-healing processor on SRAM-based FPGAs," 2014. <https://re.public.polimi.it/handle/11311/869755> [28] A. Tang, S. Sethumadhavan, S. Stolfo, "CLKSCREW," USENIX Security 2017. <https://www.usenix.org/conference/usenixsecurity17/technical-sessions/presentation/tang> [29] Y. Wang et al., "Hertzbleed," USENIX Security 2022. <https://www.usenix.org/conference/usenixsecurity22/presentation/wang-yingchen>

Background works cited from memory (verify before publishing)

[40] J. O. Kephart, D. M. Chess, "The Vision of Autonomic Computing," *IEEE Computer* 36(1), 2003. [41] J. L. Hennessy, D. A. Patterson, "A New Golden Age for Computer Architecture," *CACM* 62(2), 2019. [42] L. A. Barroso, U. Hözl, "The Case for Energy-Proportional Computing," *IEEE Computer* 40(12), 2007. [43] D. Ernst et al., "Razor: A Low-Power Pipeline Based on Circuit-Level Timing Speculation," MICRO 2003. [44] N. Dutt, A. Jantsch, S. Sarma, "Toward Smart Embedded Systems: A Self-aware System-on-Chip (SoC) Perspective," *ACM TECS* 15(2), 2016.

Model files

The financial, TCO and sustainability arithmetic in Sections 3 and 8 was run in a short Python script (NPV, IRR, payback, sweeps). Inputs are labelled as assumptions; change them before any external use.